Learning to Refer: How Scene Complexity Affects Emergent Communication in Neural Agents

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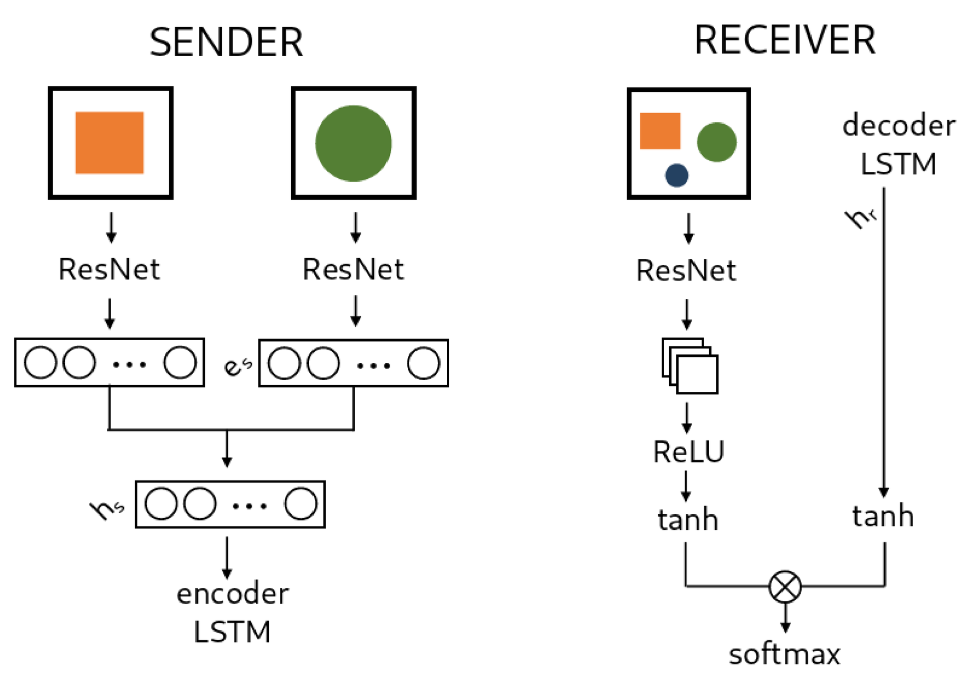
Aims

- Bridge continuous sensory input with discrete linguistic symbols
- Study symbol grounding in neural agents through interactive communication
- Investigate how scene complexity affects emergent language learning
- Examine referring expression generation in controlled visual environments
- Compare emergent vs. natural language communication protocols

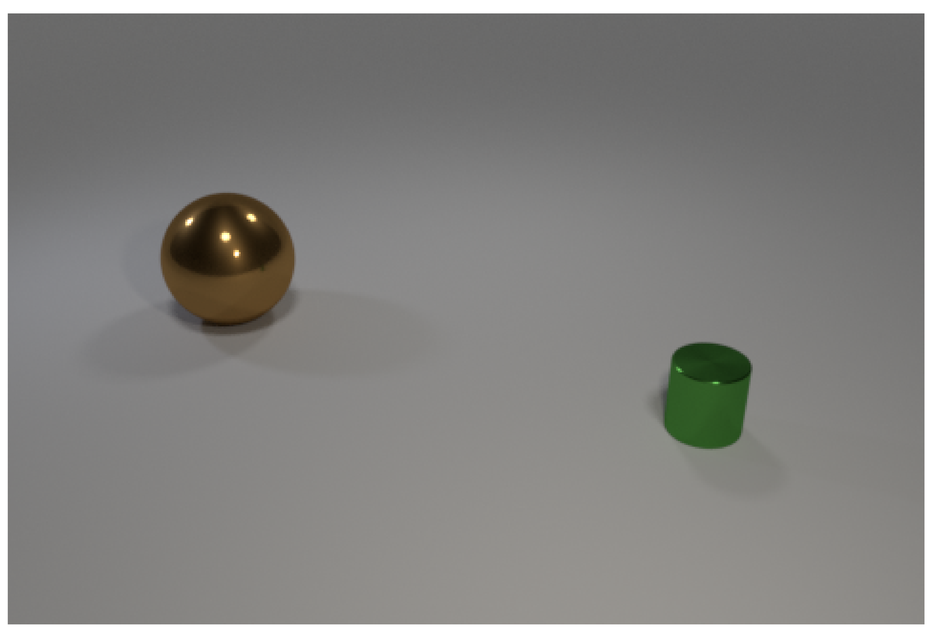
Architecture



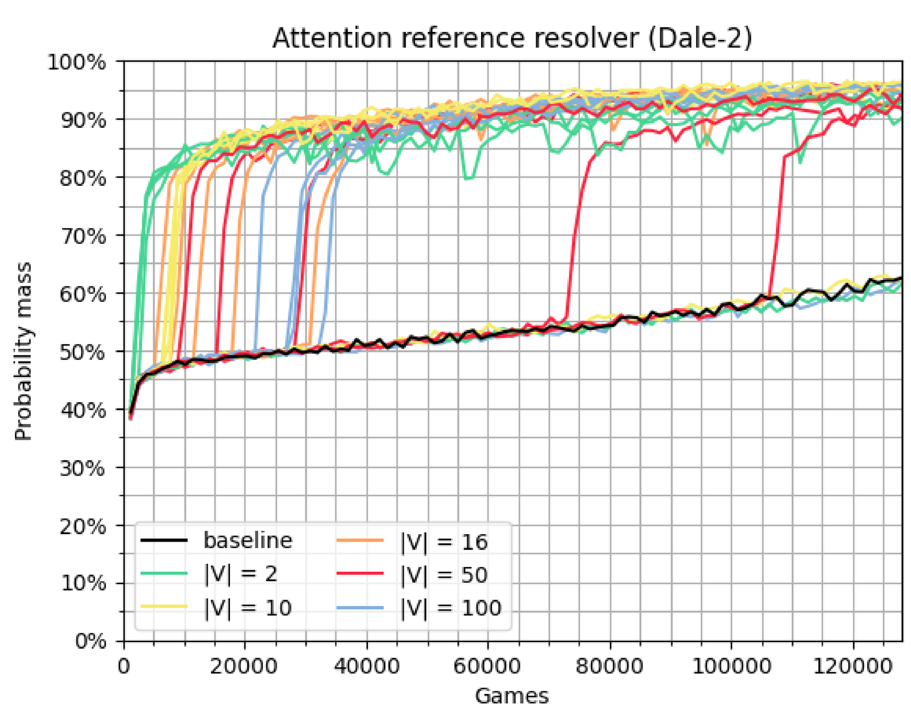
Conclusions

- Medium vocabularies (10-16 symbols) and message lengths (2-4) perform best
- Scene complexity dramatically impacts communication success
- Shape attributes easiest to learn, size attributes most challenging
- Emergent protocols achieve substantial gains over baselines
- Natural language bias improves performance significantly
- Future: Analyze linguistic properties of emergent languages

Dale-2 Dataset



Target + 1 distractor

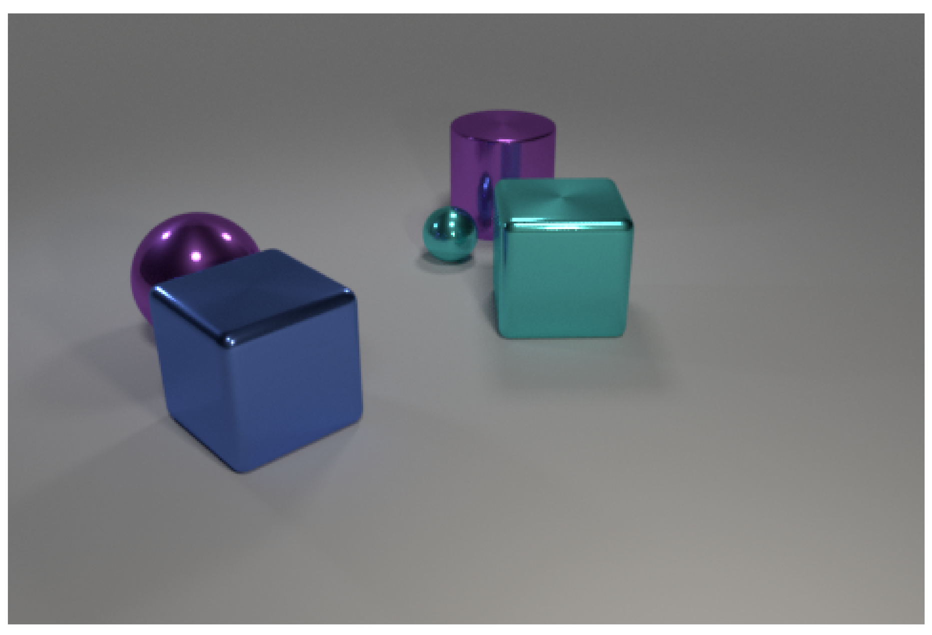


Learning curves by vocabulary size

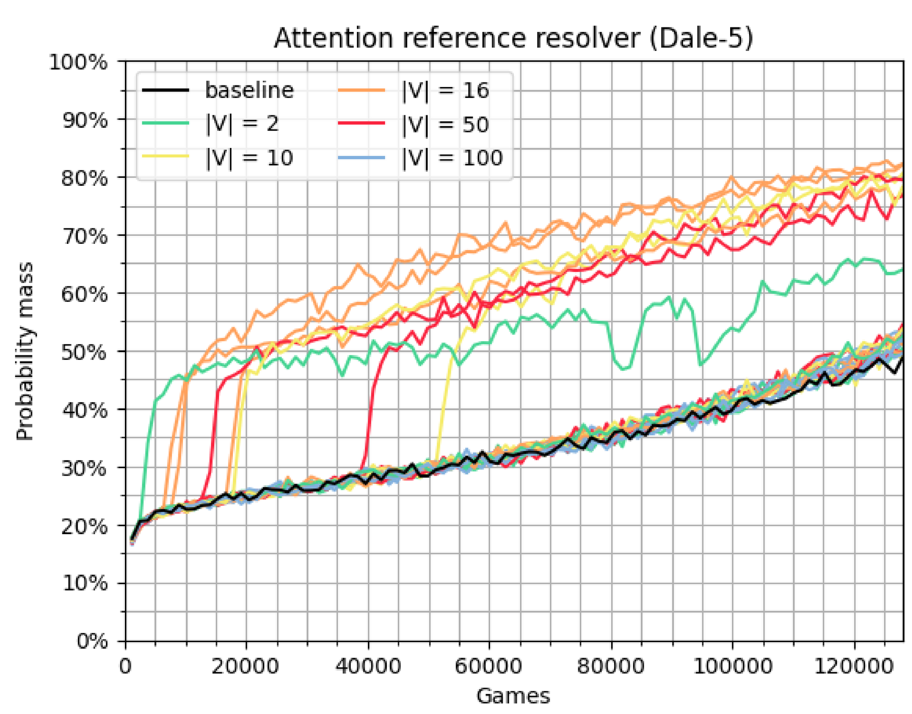
RE Generation Results:

- **Accuracy**: 99%
- **F1-Score**: 98.57%
- **Shape**: 99.75% precision
- **Color**: 98.09% precision
- **Size**: 98.73% precision

Dale-5 Dataset



Target + 4 distractors

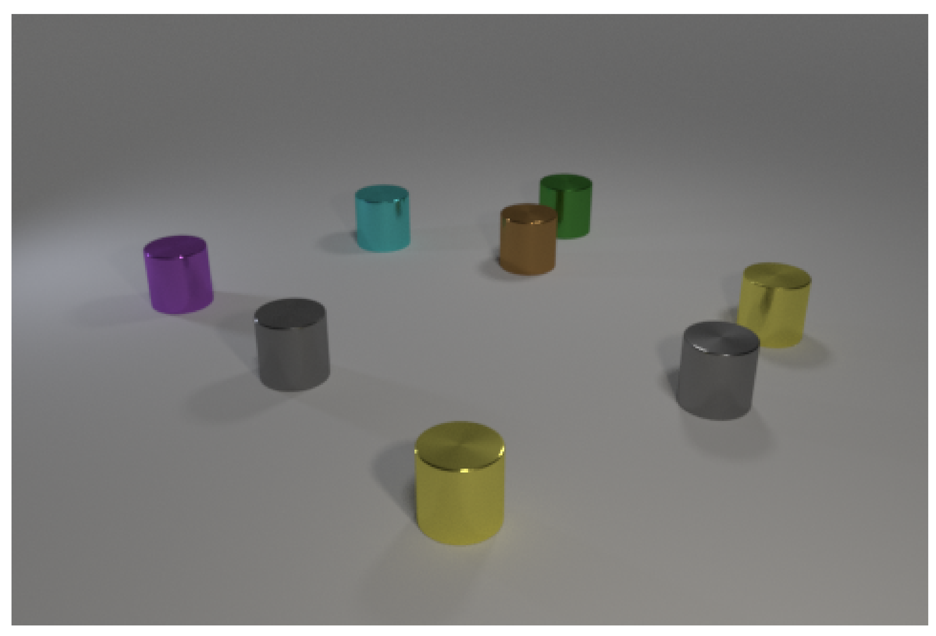


Learning curves by vocabulary size

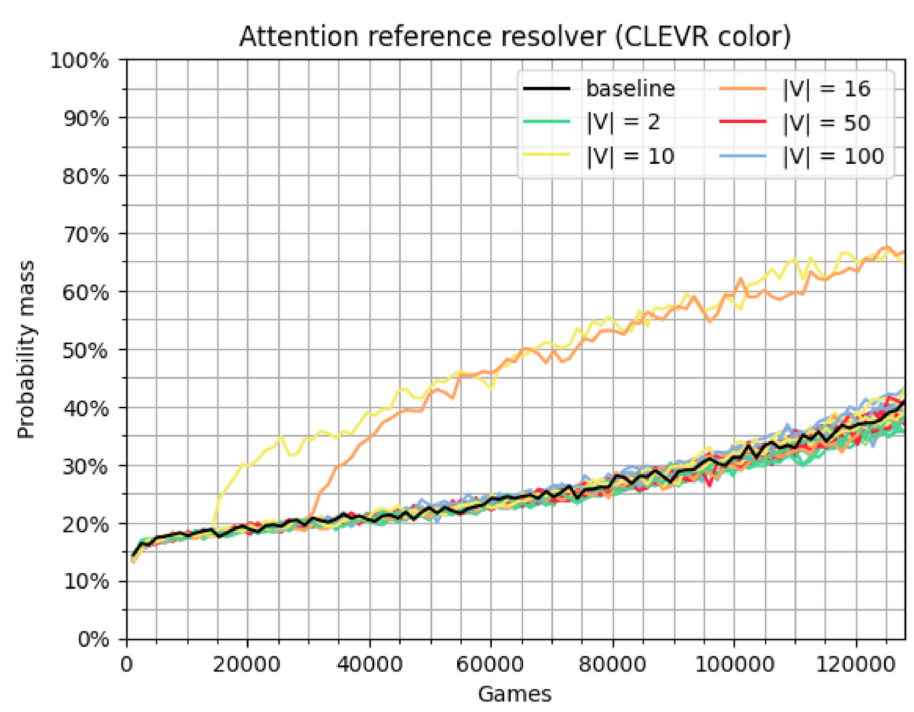
RE Generation Results:

- **Accuracy**: 69%
- **F1-Score**: 89.53%
- **Shape**: 98.30% precision
- **Color**: 92.44% precision
- **Size**: 69.43% precision

CLEVR Color Dataset



Target identified by color only



Learning curves by vocabulary size

RE Generation Results:

- **Accuracy**: 93%
- **F1-Score**: 95.17%
- **Shape**: 100% precision
- **Color**: 92.76% precision
- **Size**: N/A (not used)

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